

Smart Technology Innovations

AI-Powered Smartwatch for Health Monitoring & Security



Presented By:
Mohamed Mustafa Kamel
Abduallah Ragab
Moustafa Ahmed Elashry

The Challenges



Health Monitoring

Accurate tracking ensures effective wellness management for users.



Wearable Limitations

Current devices struggle with advanced features for users.



Security Risks

Sensitive health data requires robust protection measures.



Cost of Biometrics

Affordable devices often lack essential security features.

Key Features of Our Solution



Security

Uses DeepFace via ESP32-CAM for authentication.

Health

Continuous HR, SpO₂, and step counting provided.

Real-Time

Handles sensor fusion directly on the wrist.

Low-Cost

Built on ESP32 and ST7735 TFT display.

Core Sensors: Health & Activity Monitoring



MAX30102 Sensor

Measures heart rate and blood oxygen levels.



MPU6050

Detects motion, steps, and hand movements accurately.



TFT ST7735 LCD

Small display shows metrics and visual alerts.

Core Components

Key elements for system functionality and security

ESP32-WROOM-32

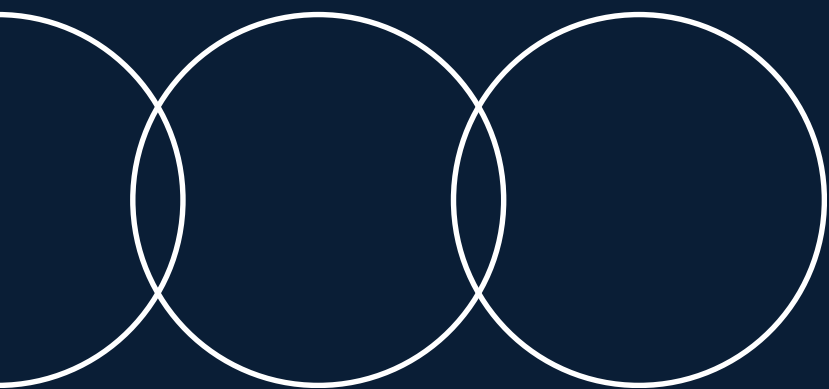
The main microcontroller for processing sensor data and managing the display, enabling efficient connectivity for various tasks.

ESP32-CAM

Captures and streams real-time video frames to the processing unit, integrating biometric authentication for enhanced security measures.

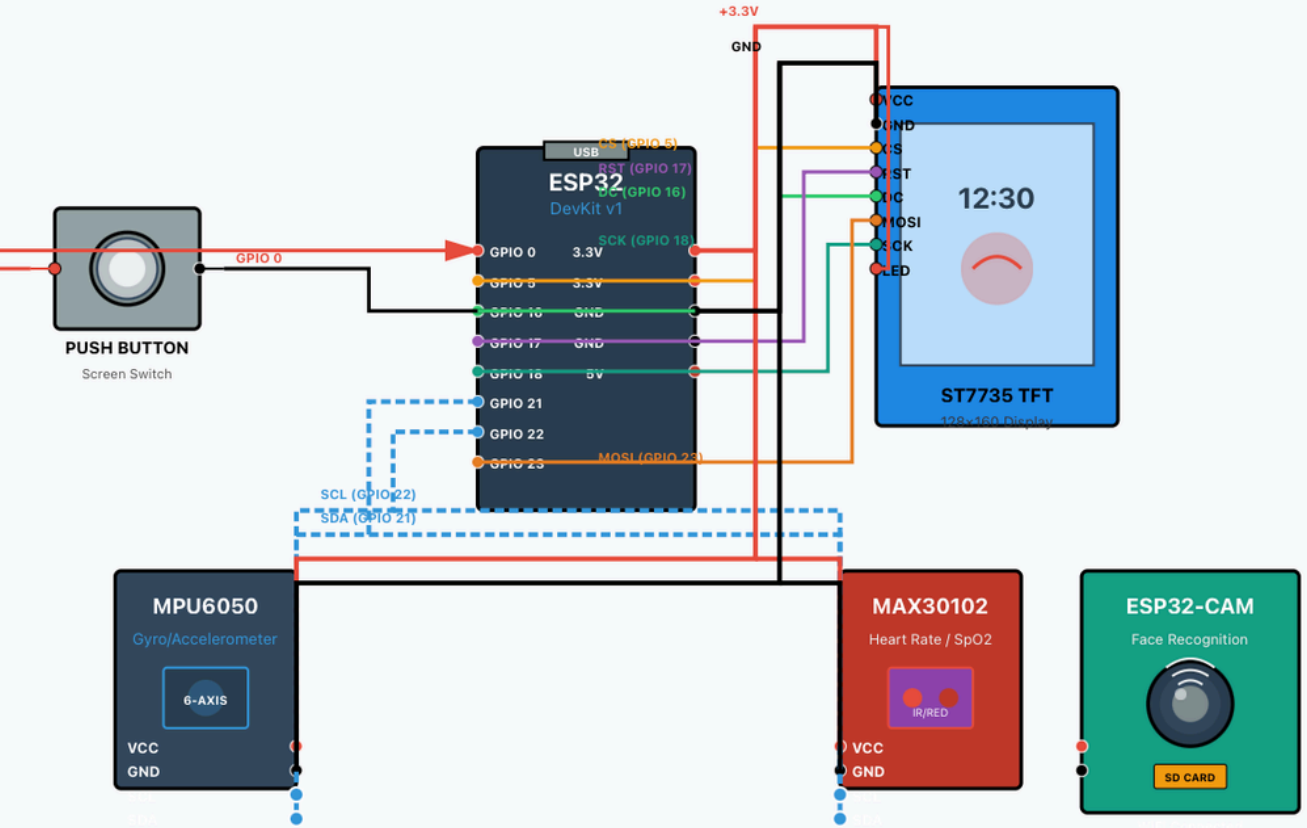
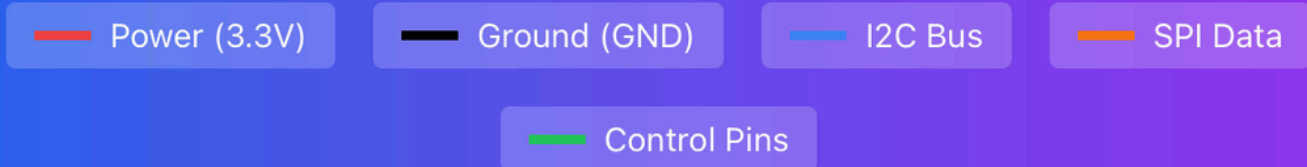
Push Button (GPIO)

Facilitates user input for cycling through screens, manual resets, and handling emergency interrupts, ensuring responsive system control.



ESP32 Smartwatch Circuit Diagram

Complete Wiring Guide for TinkerCAD



Hardware Architecture: Smartwatch Unit

Overview of Smartwatch Wiring

The **ESP32-WROOM-32** acts as the central hub, managing all components. High-speed SPI is used for the display, while the I2C bus handles the two primary sensors.

Data Flow

1

Continuous Health Monitoring of Vital Signs



Sensors

This system utilizes MAX30102 for IR/Red readings and MPU6050 for capturing acceleration data effectively and efficiently.



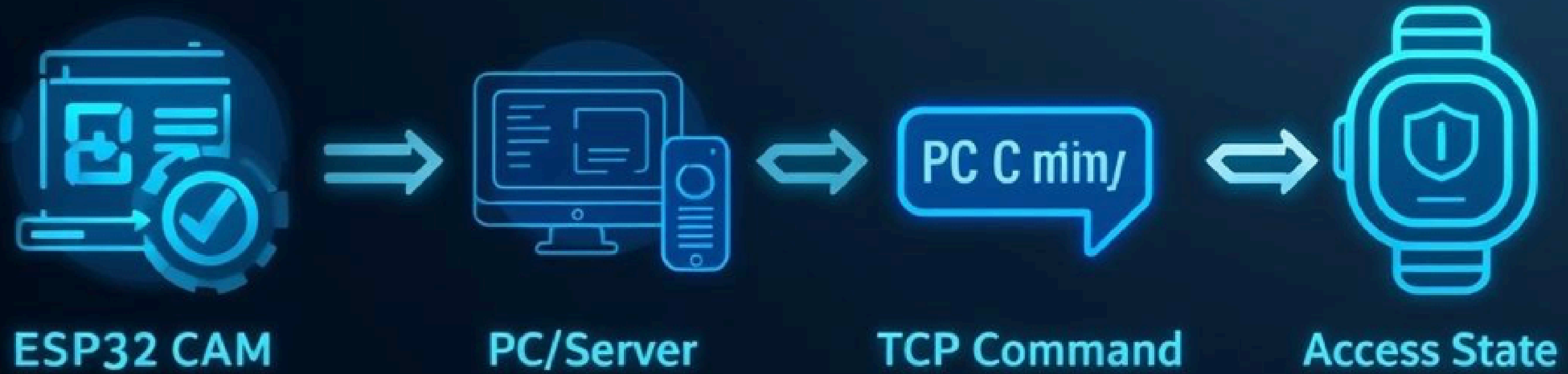
Processing

The ESP32-WROOM-32 processes signals using algorithms for heart rate and SpO₂ calculations, along with step detection through filtering techniques.



Display

The TFT ST7735 showcases real-time data on the dashboard, including heart rate, SpO₂ values, step count, dynamic graphs and also can handle notifications .



Data Flow Process 2

Face Capture

The ESP32-CAM captures the user's face, utilizing its advanced camera technology to ensure high-resolution images for accurate recognition and secure unlocking of the smart watch.

Data Transmission

The captured image is sent to the DeepFace PC via a reliable TCP connection, ensuring swift and secure transmission of data for processing and authentication purposes.

Authentication Command

Once processed, the DeepFace PC sends an "ok" command back to the ESP32, indicating successful face recognition and enabling the user to access their smart watch seamlessly.

Auto Lock

If the user does not interact with the watch within 60 seconds, the system automatically locks, enhancing security and ensuring personal data protection in case of unauthorized access.

User Experience

The entire unlocking process is designed for a smooth user experience, eliminating friction typically associated with entering PINs or patterns, offering convenience alongside robust biometric security.

Project Summary & Value Proposition

- ## 1 AI-Enhanced Security

 - ✓ First low-cost wearable integrating external face recognition for zero-input locking and unlocking.
- ## 2 Comprehensive Health Tracking

 - 🔒 Provides accurate, real-time metrics for heart health, blood oxygen, and activity levels.
- ## 3 Embedded System Efficiency

 - 🔋 Utilizes the ESP32 platform for a low-power, compact, and wireless solution.



**thank
you**